COMP 3512
Assignment #2: Single-Page App in JavaScript

Mar 26, 2026

Due Saturday April 11, 2026 at 11:59 pm

Overview

This assignment provides an opportunity for you to demonstrate your ability to work with JavaScript. The application you will be creating is a Single-Page Application to view products.

You can work in pairs (no more than two students in one group) or by yourself. The assignment is very doable by one person, but it will be a more pleasant experience sharing the work with another.

Beginning

It is your responsibility to read all the assignment instructions thoroughly and carefully. If you are unclear about something, ask me.

Grading

The grade for this assignment will be broken down as follows:

Visual Design	15%
Programming Design (best programming practices) and Documentation	10%
Functionality (follows requirements)	75%

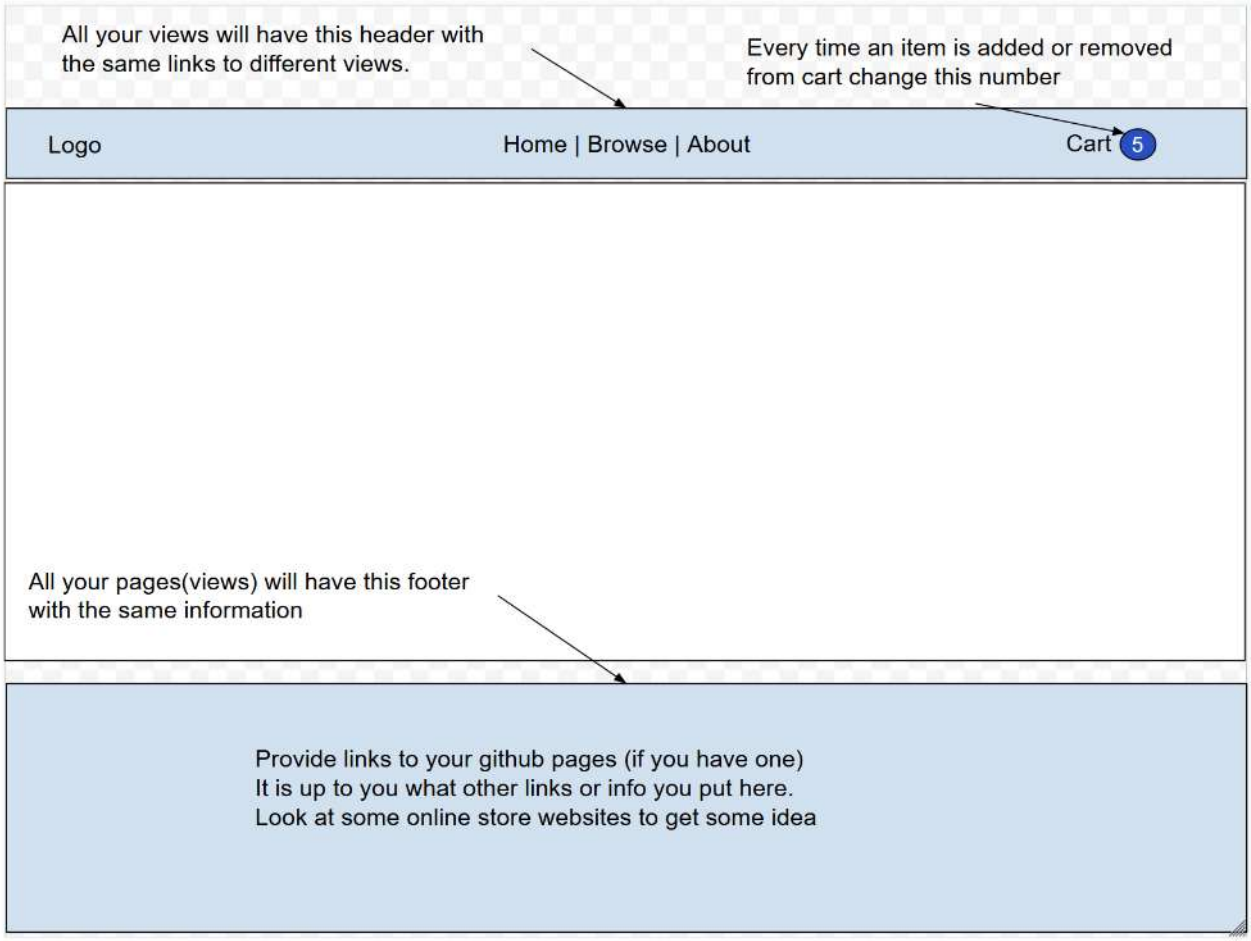
Requirements

This assignment consists of a **single** HTML page (hence single-page application or SPA) with the following functionality:

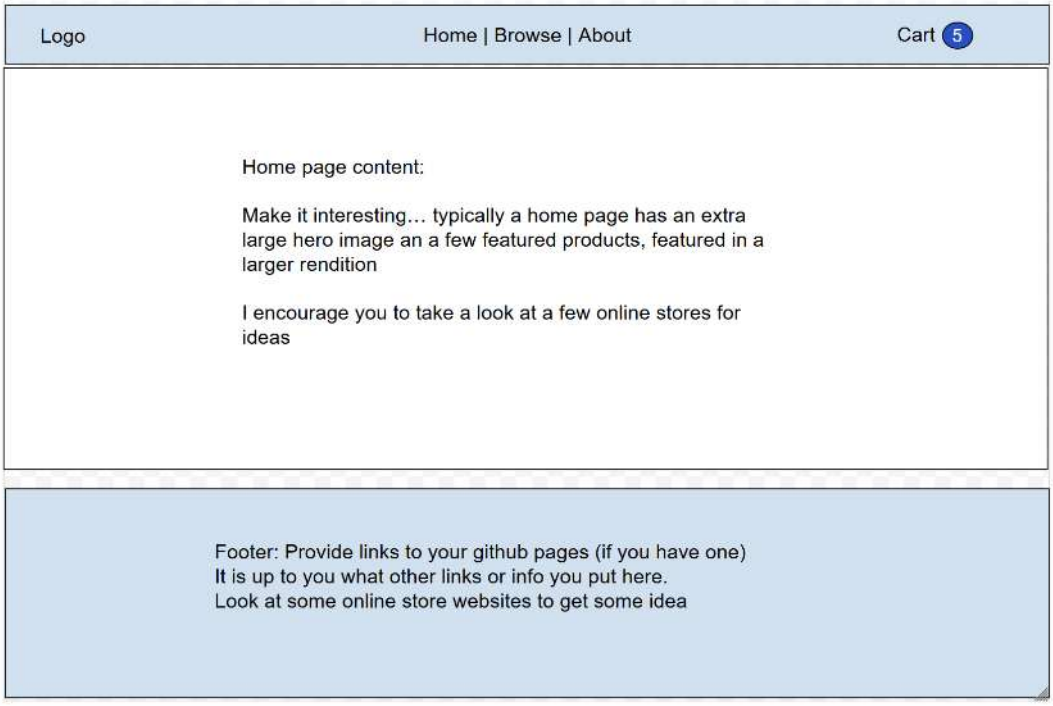
1. **You can use a third-party CSS library/framework (indeed I would encourage you to use one).** Tailwind CSS is a great thing to have on your resume, but it does have a bit of a learning curve. Be careful though that your CSS library is not using/requiring its own JavaScript library as well. Some popular CSS libraries (e.g., Bootstrap) include their own JavaScript libraries to do fancy things with select lists, modals, etc. **You are not allowed to include these** so be sure you don't have any extra <script> tags from your CSS library!
2. You must write your project using only the topics and syntax that have been covered in class. **If you use any syntax, structures, or techniques that have not been taught in lectures, marks will be deducted.** Use of features not covered in class may also be interpreted as evidence of AI assistance which is not allowed except for CSS styling.
3. **You must only implement the functionalities explicitly requested in the assignment. Adding extra features or functionalities beyond what is required will be considered as evidence of AI assistance or plagiarism, and may result in mark deductions.**
4. Your assignment should have just a single HTML page which **must be** named index.html.
5. The assignment should have four main views (**Home, Browse, Single Product** and **Shopping Cart**) and one dialog/popup (**About Us**). When the program first starts, display the **Home** view.
6. The source of data is in a provided json file (I have provided a formatted and a compact version; you should use the compact version). More detail will be provided below.

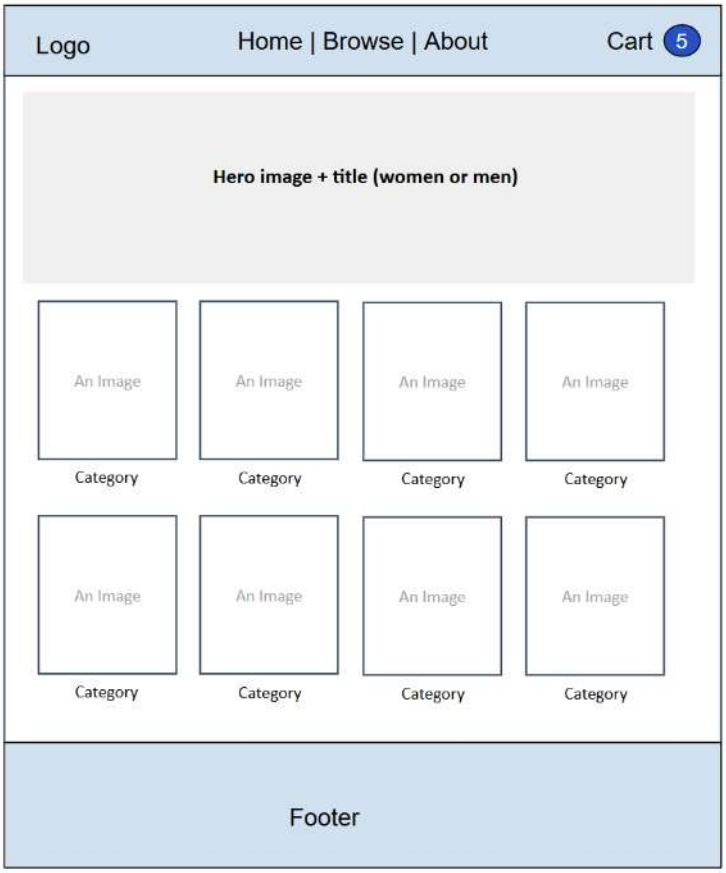
7. You must write your own JavaScript. That is, no jQuery, no Bootstrap, no other third-party JavaScript libraries. You have all the knowledge you need from the lectures and tutorials to complete this assignment. If you do find yourself, however, making use of some small snippet of code you found online (and I do mean small), then you must provide references (i.e., specify the URL where you found it) via comments within your .js file. Failure to properly credit other people's work in your JavaScript will result in a zero grade. Using antediluvian JavaScript found online will also result in lower marks. There is no need to credit code you found in the tutorials or lectures.

8. Header / Footer. All your views will contain the same header / footer. Another way of saying this is that you will be changing the content *between* the header and footer. The image below demonstrates the functionality required. The links are not URLs to other html files; instead, they indicate that your SPA must display a different view when clicked.



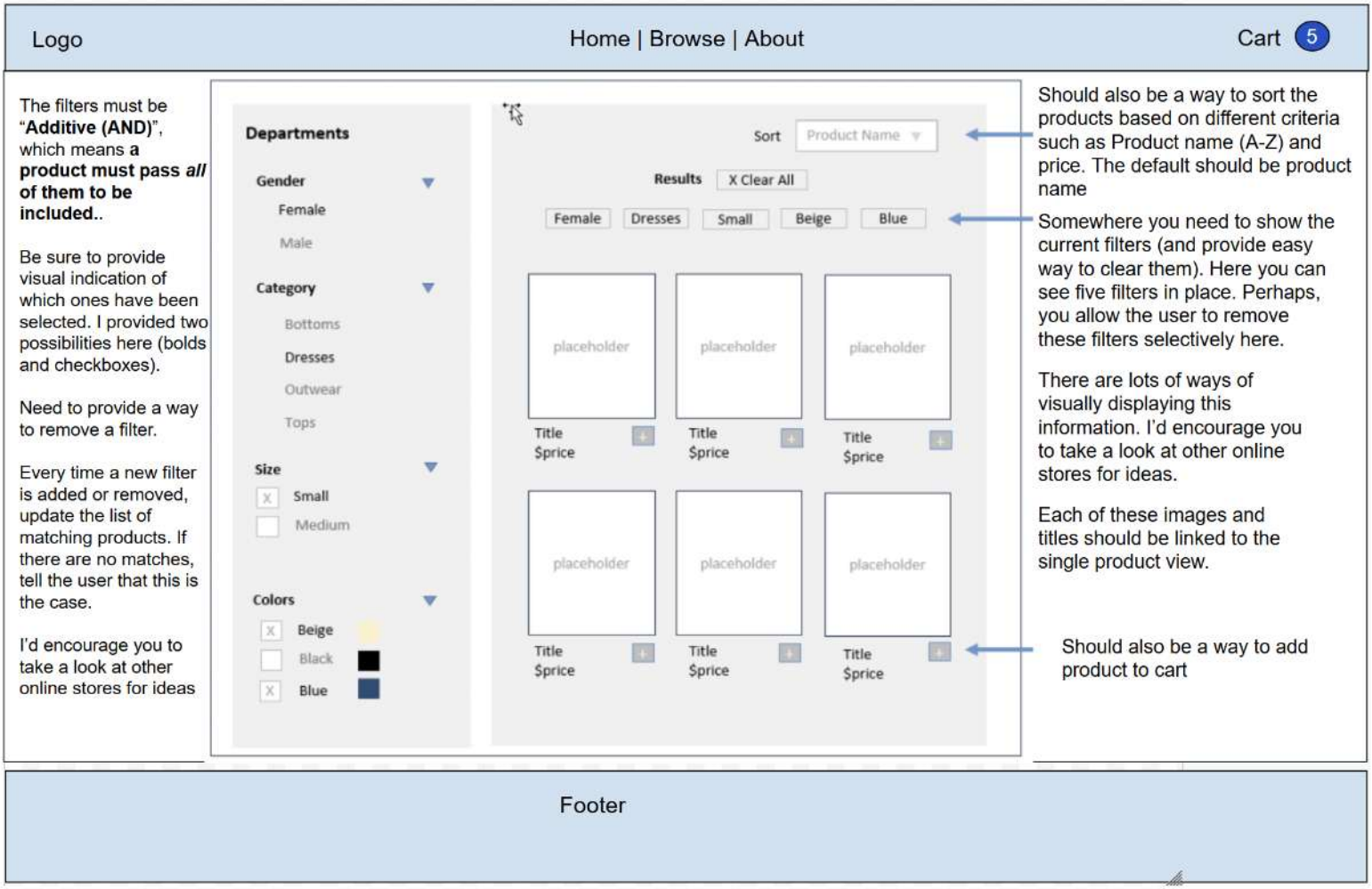
9. **Home View**. You will implement a page with similar functionality as that described below. As you can see, I've left it up to you in terms of what content to put on this page.



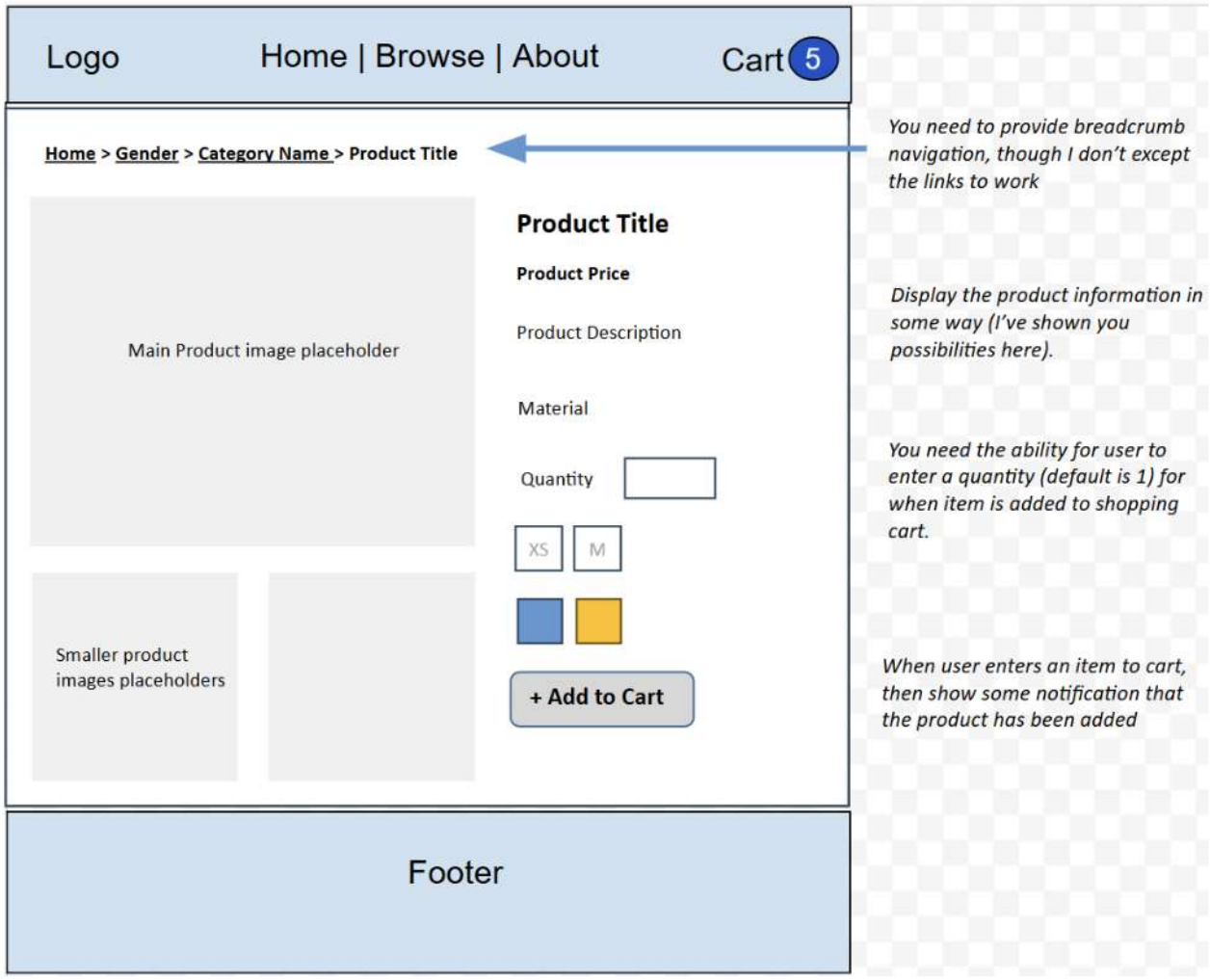


This is a rough sample idea only for home page (view).

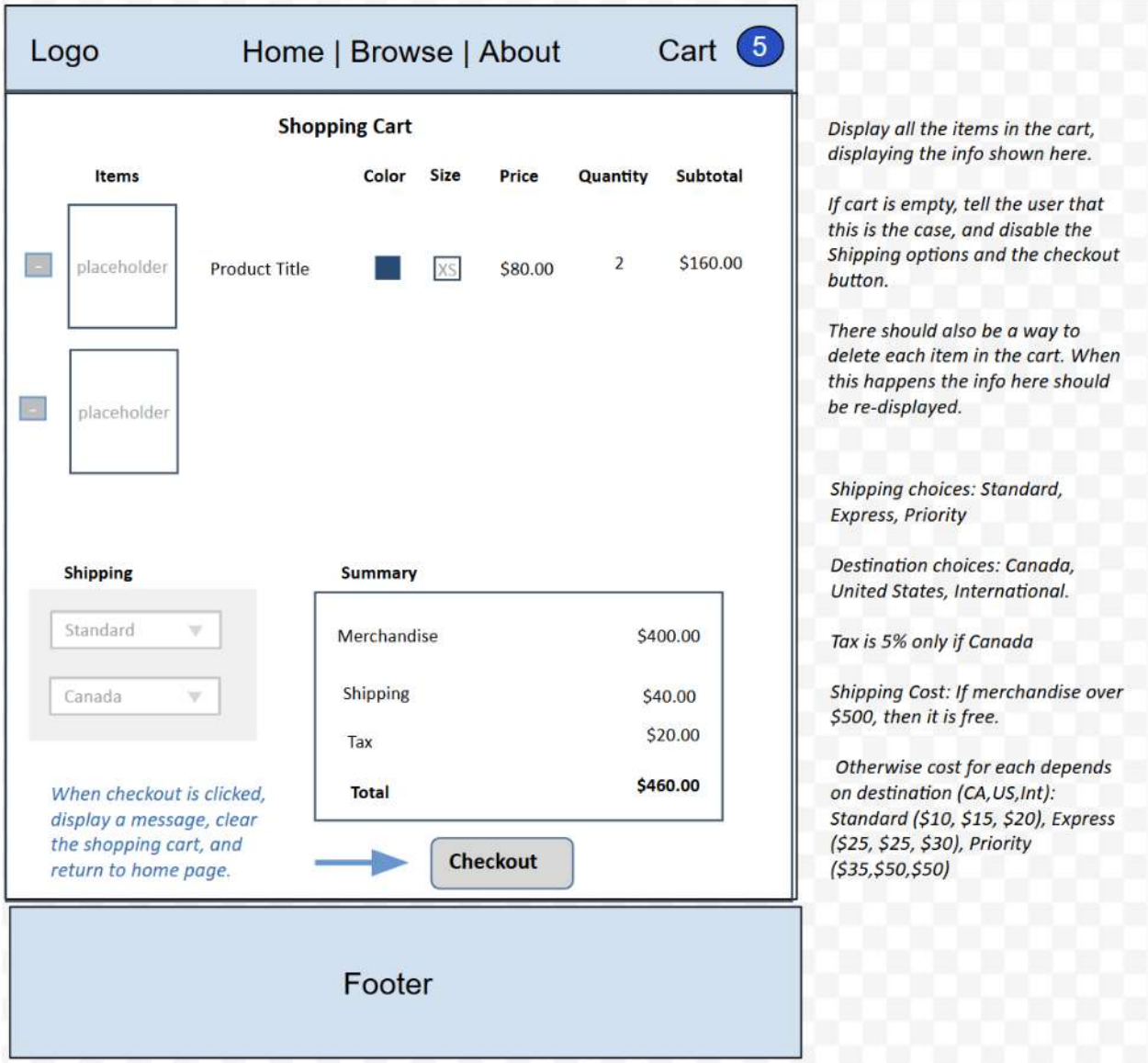
10. **Browse View.** This is the heart of the assignment. Your view must be able to display products that match a variety of different criteria and must have the functionality listed below.



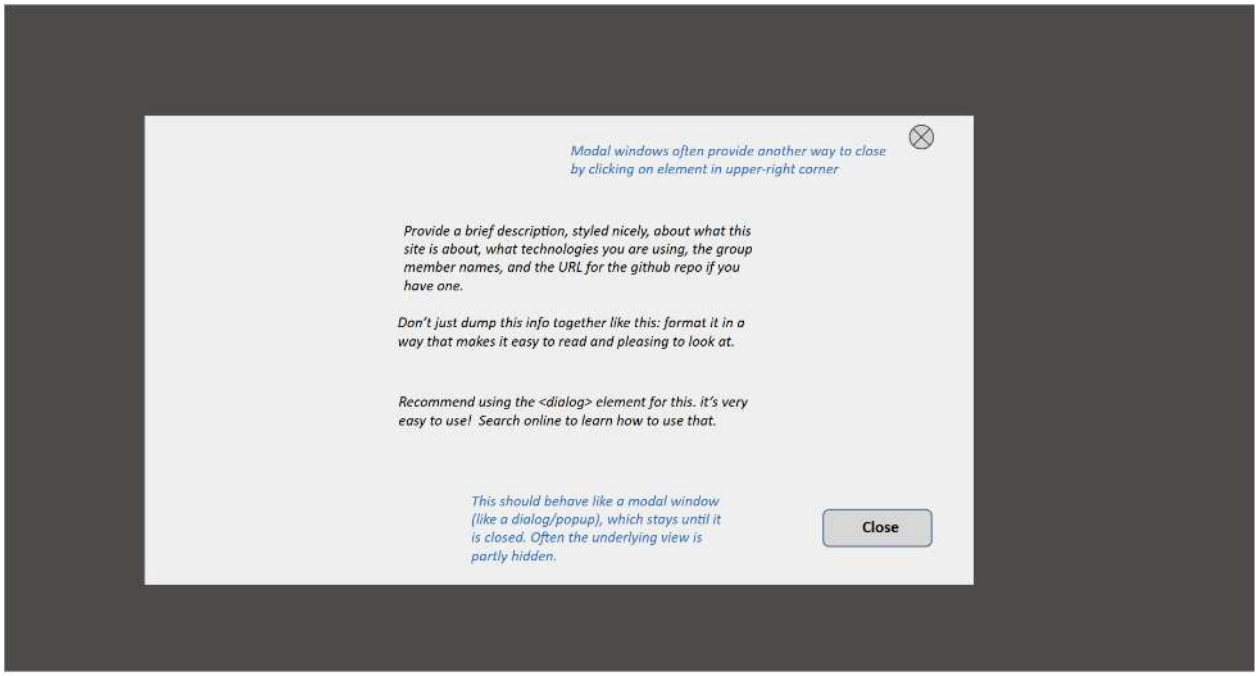
11. **Single Product View.** Display details for a single product with the following functionality.



12. **Shopping Cart View.** Display current content of shopping cart with the following functionality.



13. **About Us Dialog/Popup.** Display an about dialog that has info about the assignment, the technologies you are using, info about yourself, a link to the github repo, etc.



Handling Data

The simplest way to use JSON data is to copy and paste it inside a <script> tag, access it using DOM methods, and then use JSON.parse() method to store it in an array. Please refer to the slides for examples of these methods. You may also use more advanced methods for handling JSON data without embedding it directly in the <script> tag. Since these advanced methods have not been covered in class, please cite any resources you used to learn how to implement them. This is the only part of the assignment where you are allowed to use techniques not taught in class, if desired. However, as mentioned above, you do not need any methods beyond what has been taught in class to complete this assignment.

To see the content of this data, I have provided the data-pretty.json file for you to examine it.

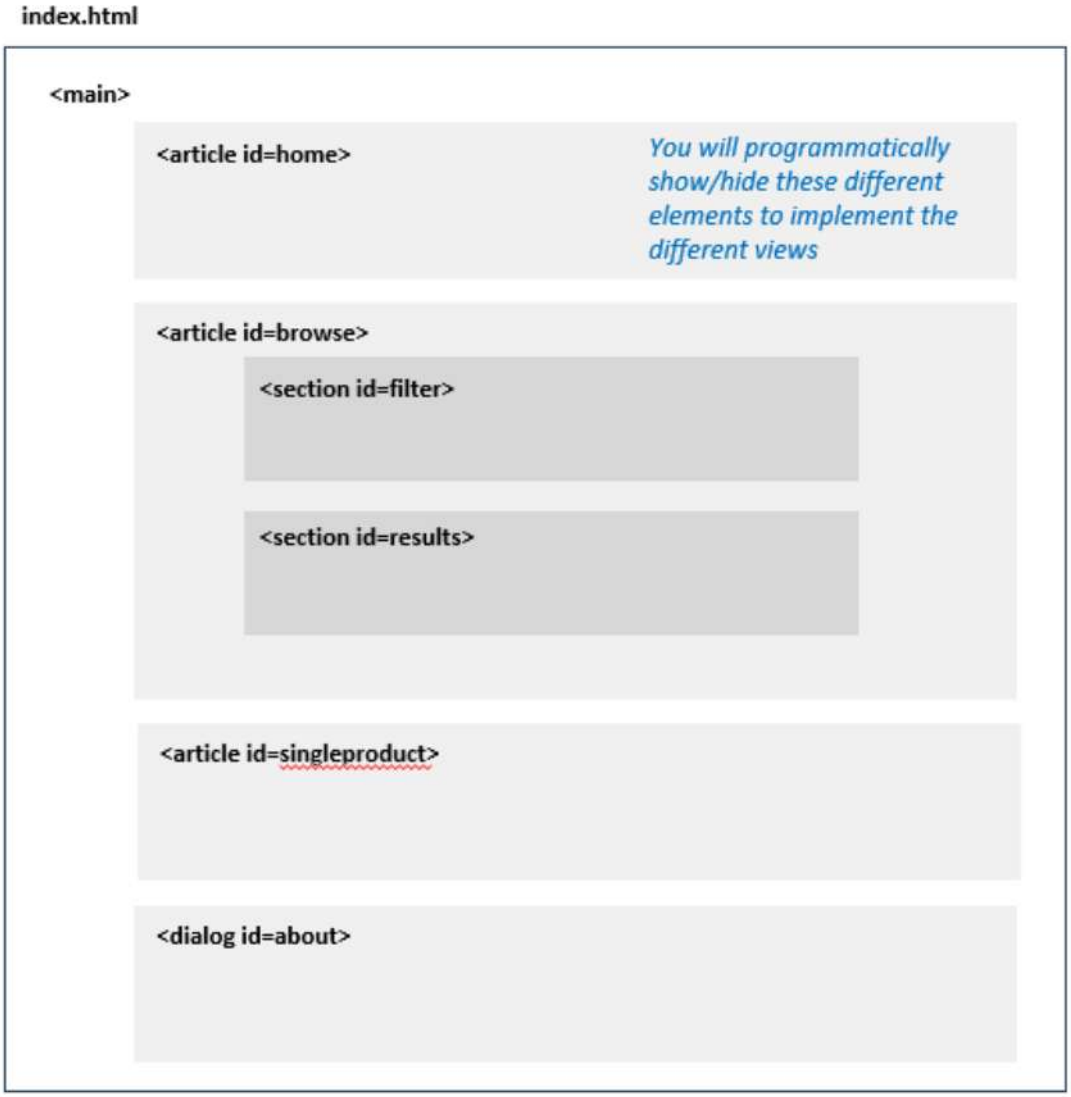
Submitting

As usual, please submit your code to the designated Assignment 1 dropbox on D2L.

Alternatively, you may share your GitHub repository with me. If you choose this option, make sure the repository is accessible and that I have the necessary permissions to view your work.

Hints

1. This assignment requires swapping different views. You should implement each view in its own container, e.g.,



2. Remember that JSON and JavaScript are case sensitive. For instance, it is easy to type in Category and it won't work because the JSON field is actually category.

3. Your index.html file will include the markup for all your views. Your JavaScript code will programmatically hide/unhide (i.e., change the display value) the relevant markup container for these views.

4. I would recommend implementing the **Single Product View** first and then the **Shopping Cart View** second. The **Browse View** is the most complex so maybe do that third (though some programmers prefer to tackle the most complex use case first).

5. Most of your visual design mark will be determined by how much effort you took to make the views look well designed. Simply throwing up the data with basic formatting will earn you minimal design marks.

6. Most years, students tend to get low marks on the design side of the assignment. I would recommend looking at other sites on the internet and examine how they present data. Notice the use of contrast (weight, color, etc) and spacing.
7. Most of your programming design mark will be based on my assessment of your code using typical code review criteria. For instance, did you modularize your code using functions? Are your functions too long? Is the code documented? Do you have a lot of code duplication (you shouldn't … if you are copy+pasting code, that should tell you that you are doing things wrong from a design standpoint)? Are variables and functions sensibly named? Are you using outdated JavaScript techniques (e.g., inline coding, var, XmlHttpRequest, etc)? Is the code excessively reliant on found code from Stack Overflow, etc?